

ROMASA QASIM

Research Assistant, Bouchnita Lab, University of Texas at El Paso

rqasim@miners.utep.edu | (915) 843-3747 | University of Texas, El Paso, 500 W. University Ave., TX, USA, 79968 | [LinkedIn](#)
| [Github](#) | [Google Scholar](#)

Summary

PhD candidate in Computational Science at UTEP with strong technical problem-solving skills. Proven experience in developing and improving computational efficiency using Python, TensorFlow, and MATLAB. Currently working on mathematical and computational models to design a treatment modeling platform for pre-clinical assessment of treatment.

Education

PhD in Computational Science	01/2022 - Present
University of Texas at El Paso (UTEP) El Paso, Texas, USA	
Research Area: Mechanistic modeling of tumor microenvironment	
Advisor: Anass Boucnita, PhD	
Expected completion date: December 2026	
MS in Computational Science	01/2022 - 05/2025
University of Texas at El Paso El Paso, Texas, USA	
Research Area: Agent-based modeling of tumor progression	
CGPA: 3.94/4.00	
MS in Computer Science & Engineering	09/2011 - 08/2013
North South University Dhaka, Bangladesh	
CGPA: 3.97/4.00 (Vice-Chancellor's Gold Medal)	
BS in Computer Science	01/2006 - 12/2009
University of Karachi Karachi, Pakistan	
CGPA: 3.28/4.00	

Publications

Journal Articles:
1. Romasa Qasim, Anass Boucnita (2026). From 3D stochastic multiscale simulations to population dynamics models: the impact of EGFR mutations on tumor initiation. [Submitted]
2. Romasa Qasim, N. Kauser, & Tahseen Jilani (2010). Secondary structure prediction of RNA using machine learning method. International Journal of Computer Applications (IJCA), 10(6), 15–22.
Theses:
1. Romasa Qasim (2025). Multiscale integration of receptor-ligand dynamics into discrete & continuous tumor growth models with application to tyrosine kinase inhibitor treatment. (Masters Thesis, Computational Science Program, University of Texas, El Paso).

Presentations

1. Romasa Qasim, Anass Boucnita, "Multiscale models of tumor growth with explicit receptor-ligand dynamics to quantify tumorigenesis conditions", 32nd Joint UTEP/NMSU workshop on Mathematics, Computer Science, & Computational Sciences, El Paso, TX, USA, Nov 2025
2. Romasa Qasim, Anass Boucnita, "Multiscale, continuous and data-driven surrogate models to analyze the conditions influencing tumor growth during EGFR overexpression", Institute of Biological Engineering Annual Conference 2025, Salt Lake City, Utah, Sep 2025.
3. Romasa Qsim, Anass Boucnita, 'A 3D Multiscale Model to Explore the Role of EGFR Overexpression in Tumorigenesis, 30th Joint UTEP/NMSU Workshop on Mathematics, Computer Science, and Computational Sciences, El Paso, TX, USA, Nov 2023

Experience

Research Assistant, Bouchnita Lab	01/2024 - Present
The University of Texas at El Paso Texas, USA	
- Developed fine-grained model to capture receptor-ligand interactions at physiological levels - Extended the 3D agent-based multiscale model to incorporate detailed tumor microenvironment, focusing on receptor-ligand dynamics and by improving computational efficiency by 25% by using differential equations via numerical methods - Modeled the pharmacokinetics and pharmacodynamics profiles of tyrosine kinase inhibitors targeting receptor to evaluate therapeutic efficacy, that were presented at a biannual University of Texas, El Paso conference - Implementing Monte Carlo simulation and mathematics-informed neural network for the synthetic dataset to predict tumor state	

Graduate Teaching Assistant

University of Texas at El Paso | Texas, USA

- **Facilitated** weekly discussion sessions for 100+ undergraduate students in Calculus I & II
- **Evaluated** and provided detailed, constructive feedback on exams, lab reports, and assignments for classes of up to 30 students
- **Delivered** substitute lectures on Calculus I for undergraduate courses

09/2022 - 12/2023

Lecturer

Independent University, Bangladesh | Dhaka, Bangladesh

- **Delivered** comprehensive lectures and led interactive laboratory sessions for 3 undergraduate courses per semester
- **Evaluated** student performance through the design and grading of complex assignments, exams, and technical projects, providing targeted feedback to improve student outcomes
- **Designed** and implemented new syllabi, coursework, and laboratory modules for Introduction to Computer Science course by replacing traditional C++ with Python programming language
- **Guided** 16 undergraduate and graduate students to successfully complete their internship at various software houses

08/2017 - 07/2022

Projects

Zeus: XGBoost recommendation system for hypertension patitents

MediTrak Life | El Paso, Texas (Online)

- *Engineered a transfer learning framework in Python using XGBoost to recommend daily activities to hypertension patients, resulting in an 8% improvement in next day's blood pressure accuracy over baseline Random Forest models*
- *Deployed pre-trained parameters of 300+ volunteers from the MyHeartCounts dataset to analyze Apple Watch data*
- *The project won 3rd position in the Elevate HealthTech Challenge 2024.*

01/2024 - 03/2024

Cell quantification using computer vision and machine learning

in collaboration with Univeristy of Texas, San Antonio | El Paso, Texas (Online)

- *Developed a machine learning pipeline for cell segmentation to improve performance by reducing relative error rates from 16% to 8% compared to baseline CellPose model*

11/2025 - 08/2026

Internships

Software Developer Intern

North South University | Dhaka, Bangladehs

- *Engineered a full-stack, paperless Human Resource and Accounting application Symphony 1.4 framework, replacing legacy manual systems for the university's administrative departments*
- *Architected the SQLServer database from scratch and developed core backend modules to automate salary generation and administrative operations for over 300 staff members*

01/2013 - 07/2013

Skills

Technial Skills: Medical image analysis, analytical reasoning, mathematical modeling, automation solutions,
Scientific Tools: Tensorflow, Keras, PyTorch, MATLAB, Anaconda, LINUX, Git, Docker, and ParaView
Programming Languages: Python, R, C++, C#, C, SQL scripting, Java, and Shell Scripting, Node.Js,
Soft Skills: Team-work, leadership, adaptability, emotional intelligence, communication & resilience
General Computing: Windows 11, Linux, Office 365, iPhone/Andriod, Eclipse IDE

Certifications

Fully funded Summer camp on Data and AI Intensive Research with Rigor and Reproducibility (DAIR³)

University of Texas at San Antonio (UTSA) | San Antonio, Texas, USA

Hands-on machine learning and HPC workshop

University of Texas at El Paso | El Paso, USA

06/2025 - 06/2025

09/2025 - 10/2025